Architecture & Design Notes

The platform is designed as an end-to-end retail inventory intelligence pipeline. Raw sales and inventory data are ingested into Databricks, transformed through a Medallion architecture, and converted into operational inventory-risk and replenishment decisions.

1. Data Flow

Source: Mendeley Retail Transactions and Stocks Data — separate sales and inventory CSV files.

Bronze: Source tables preserve the incoming structure and provide a reproducible starting point for downstream processing.

Silver: Columns are standardized, dates and inventory validity periods are cleaned, returns are represented correctly, and data-quality checks are performed.

Gold: Business-ready tables are produced at useful analytical grains: daily product-store demand, latest inventory risk, and replenishment recommendations.

2. Gold Data Products

Table	Grain	Purpose
gold_daily_demand	Product × Store × Date	Daily demand, sales, margin and rolling demand metrics
gold_inventory_risk	Product × Store	Latest inventory position, coverage and risk classification
gold_replenishment_recommendations	Product × Store	Actionable target stock and recommended order quantity

3. Inventory Risk Logic

Inventory coverage is calculated as quantity on hand divided by the 30-day average daily demand when valid. Negative inventory is treated as an inventory deficit. Zero stock with positive recent demand is classified as stockout risk. Missing inventory history and internal inventory gaps remain explicitly unknown.

4. Replenishment Logic

Target stock: ceil(average daily demand over 30 days × 14). The 14-day target is a documented business assumption representing a planning horizon plus buffer; it is not a fact supplied by the dataset.

Recommended order: max(target stock – quantity on hand, 0). For negative stock, this naturally covers the deficit before bringing inventory back to the target level.

5. Technology Stack

Databricks Free Edition • PySpark • Spark SQL • Delta tables • Databricks SQL • Databricks Dashboards • Medallion Architecture

6. Dashboard Outputs

The dashboard provides KPI cards, inventory-risk distribution, monthly demand trend, stockout risk by store, top replenishment priorities, store risk scoring, recommended replenishment units by store, and a concise business-rule summary.

Retail Inventory Intelligence

System Architecture

Source Data (Mendeley Retail Dataset)

Products
product details, category, cost, price

Stores
store details, location, type

Transactions
daily sales at product-store level

Inventory
current stock, inventory dates

Databricks

Medallion Architecture

Bronze Layer

Raw Ingestion

Ingest CSV files from Mendeley and store raw data in Delta Lake

Silver Layer

Cleaned & Standardized

Data cleaning, standardization, deduplication and business rule validations

Gold Layer

Business-Ready

Create analytical tables with demand metrics, inventory position, risk classification and replenishment recommendations

Inventory Intelligence & Decision Logic

Inventory Risk Classification
Stockout, Deficit, Critical, High, Medium/Low

Replenishment Logic
Recommended order quantity based on demand and target stock

Store Risk Scoring
Aggregate store-level risk and prioritize high-risk stores

BI Dashboard (Databricks SQL)

Key Metrics, Inventory Risk Analysis, Demand Trends, Top Replenishment Priorities and Store Risk Analysis